

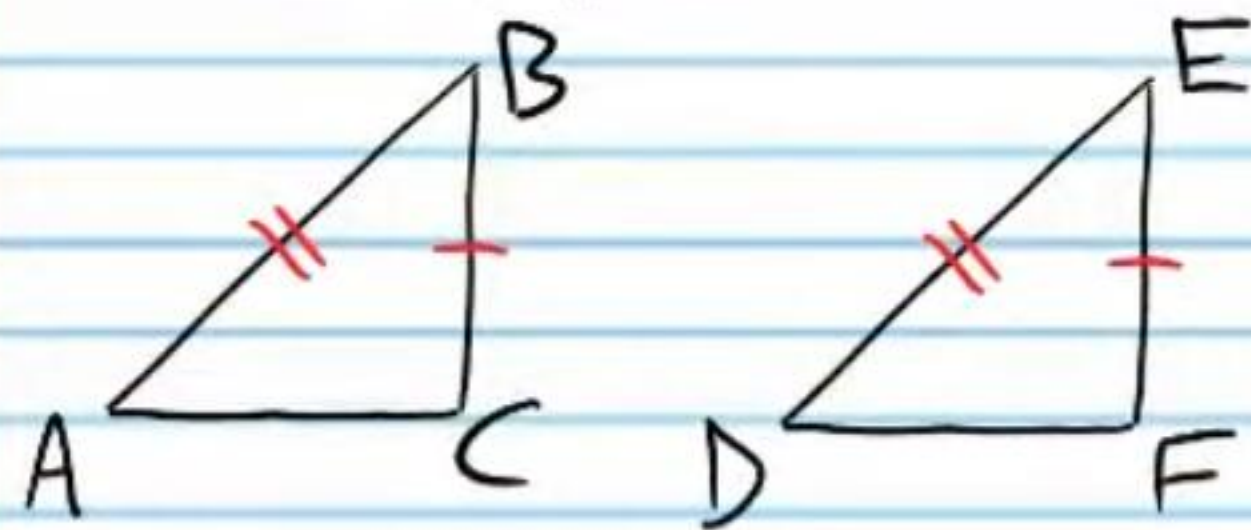
Practice Geometry Test II (Version 2)

Name
Date
Course

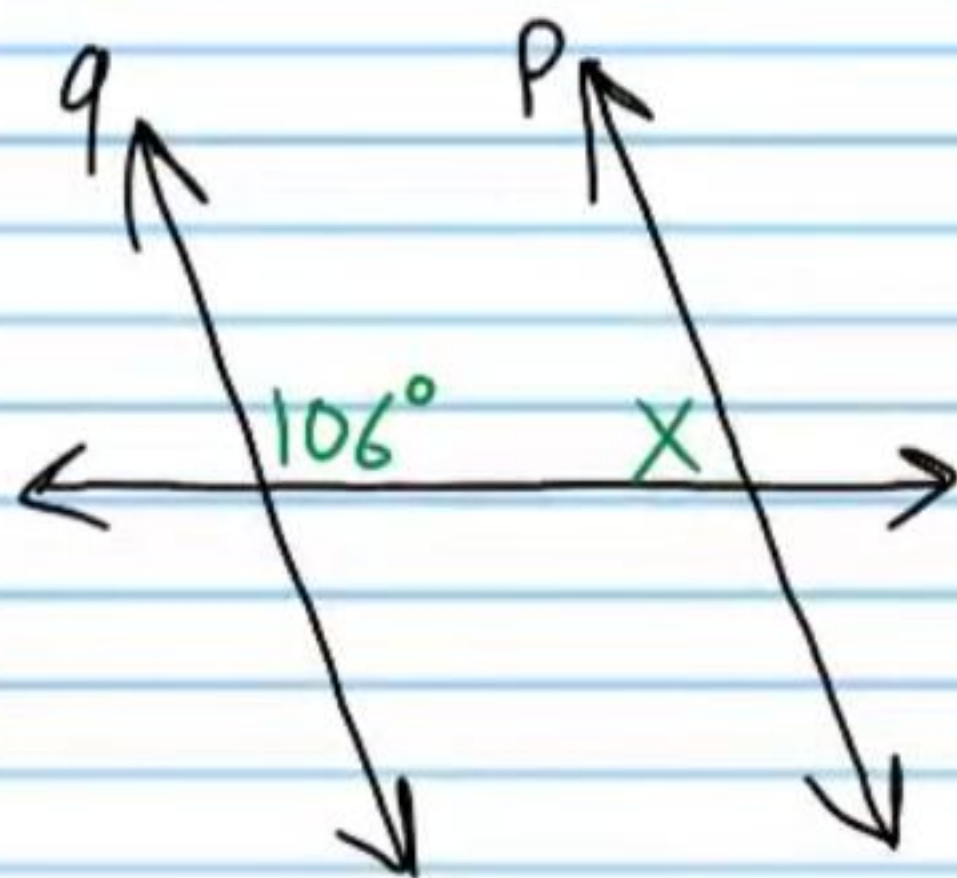
Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

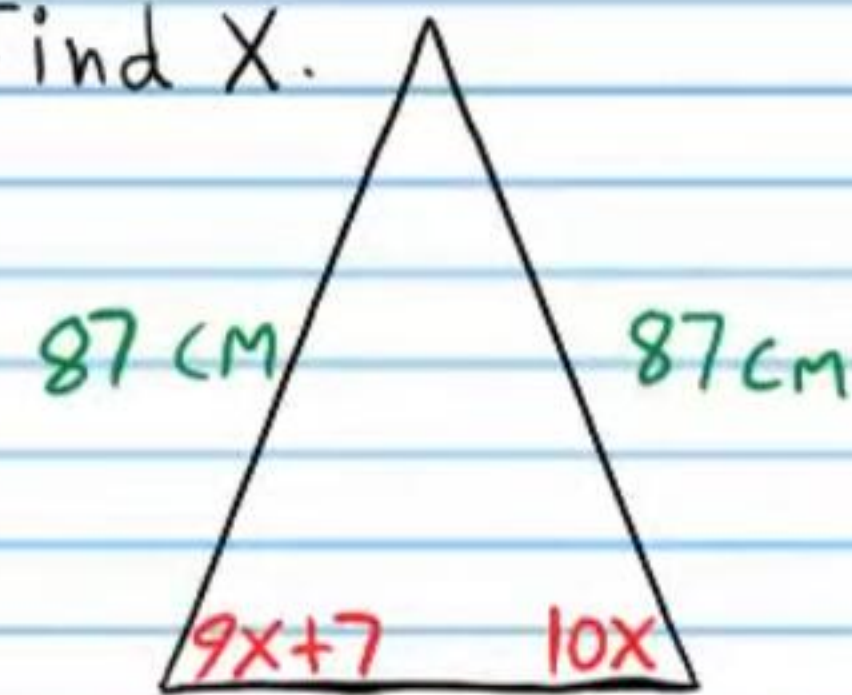
① Are the triangles below congruent? If so, write a proper congruence statement and the theorem or postulate used.



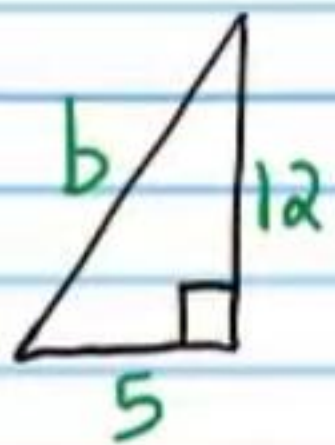
② Find x such that $q \parallel p$ and state the theorem or postulate used. You do not have to write a two-column proof but you must show the equation used to find x .



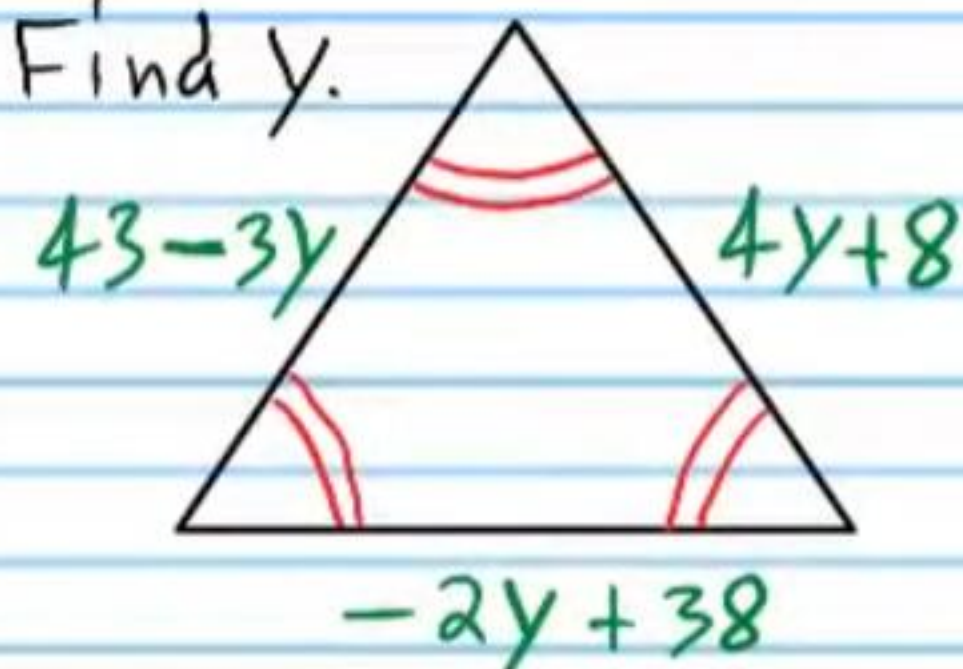
③ Find x .



④ Find b .

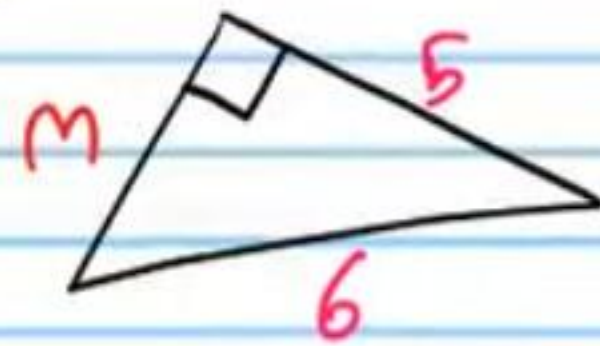


⑤ Find y .

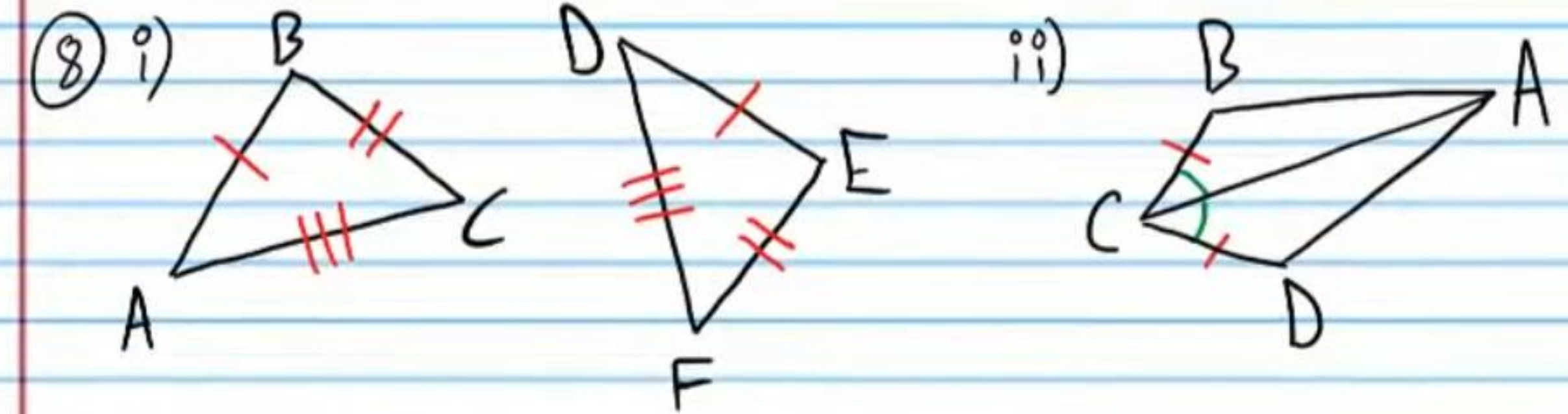


⑥ If $(10,0)$ is the midpoint between points A and B, and the coordinates of B are $(1000,3)$, find the coordinates of point A.

⑦ Find m .

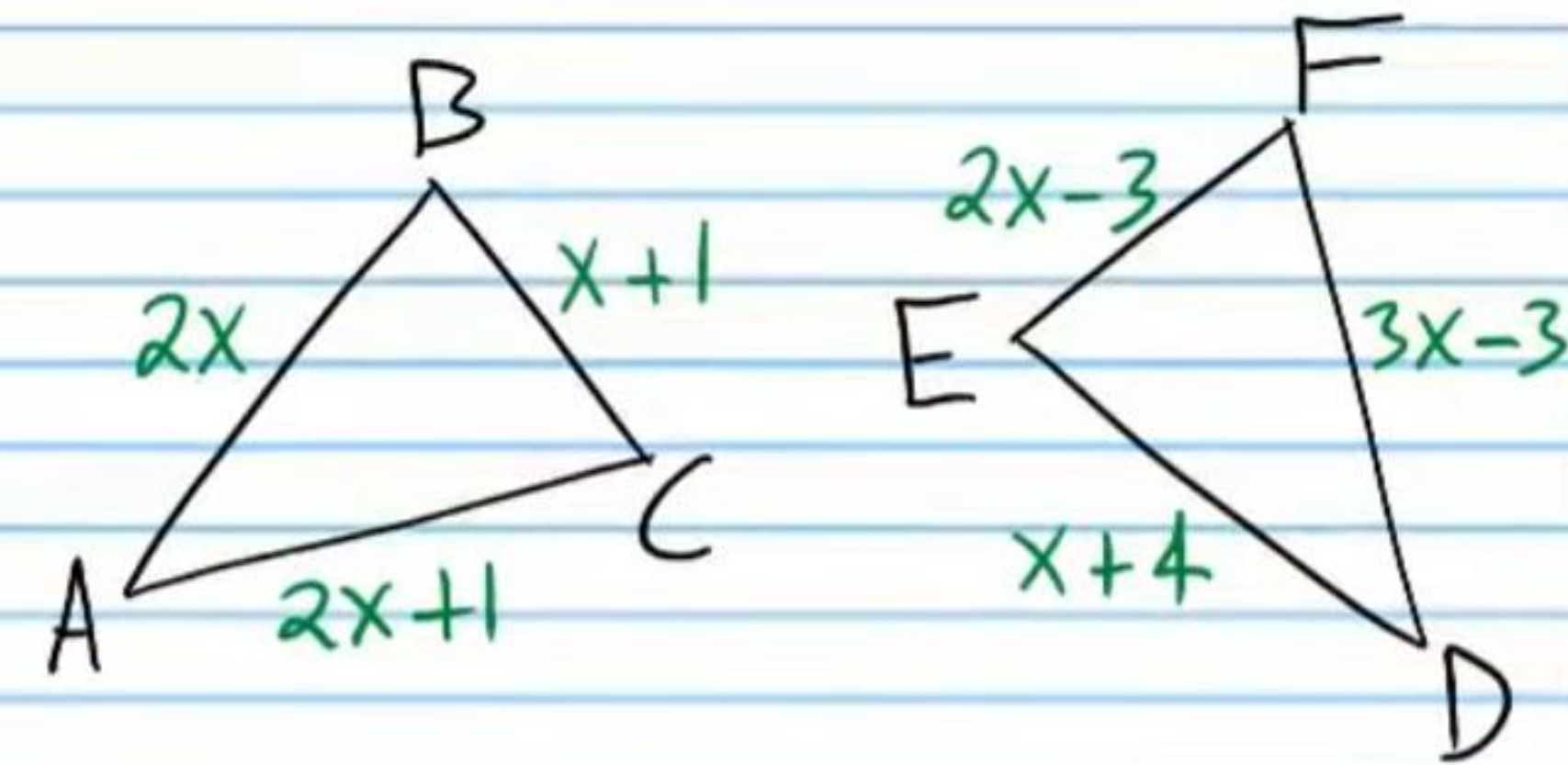


In problem #'s 8-10 below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).



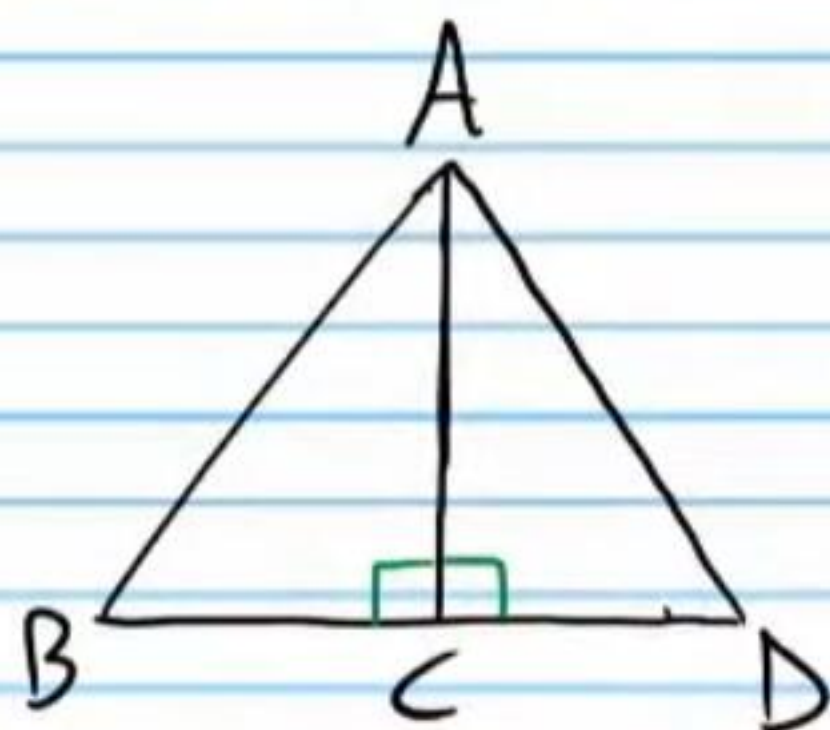
$\triangle ABC \cong$ _____

⑨ i) $x = 4$



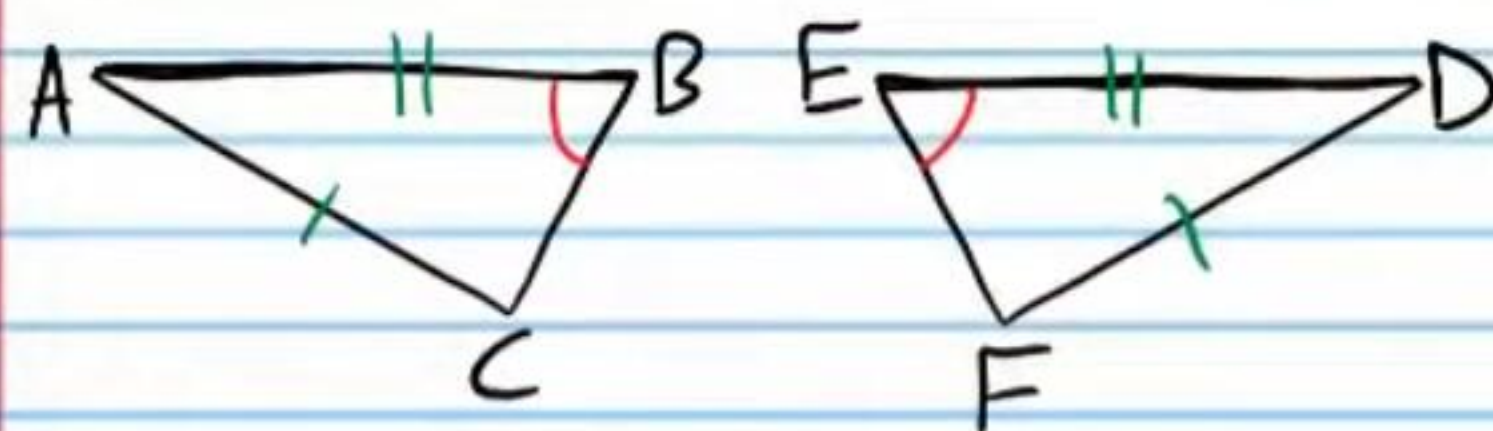
$\triangle ABC \cong$ _____

ii) $\angle BAC \cong \angle DAC$

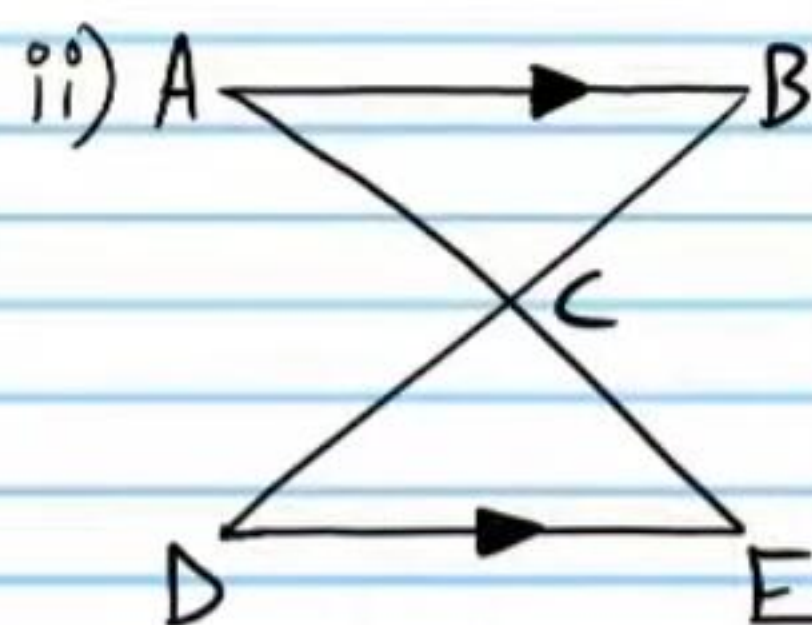


$\triangle ABC \cong$ _____

(10) i)

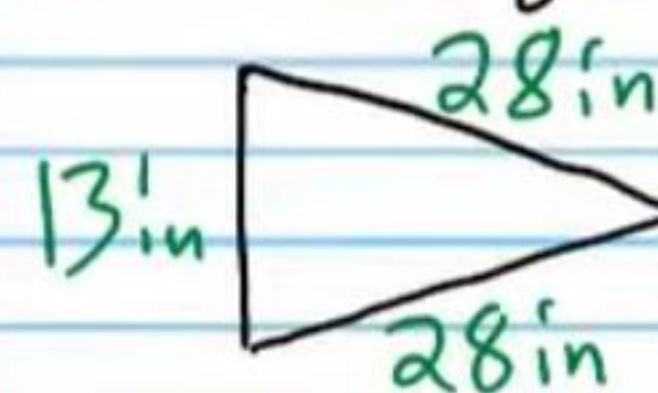


$\triangle ABC \cong$ _____

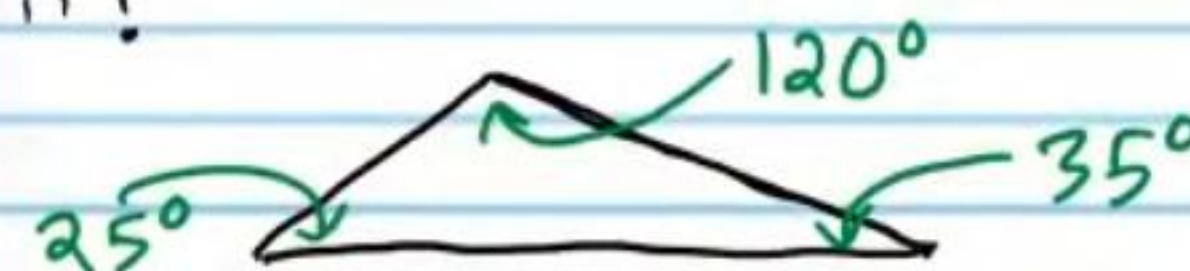


$\triangle ABC \cong$ _____

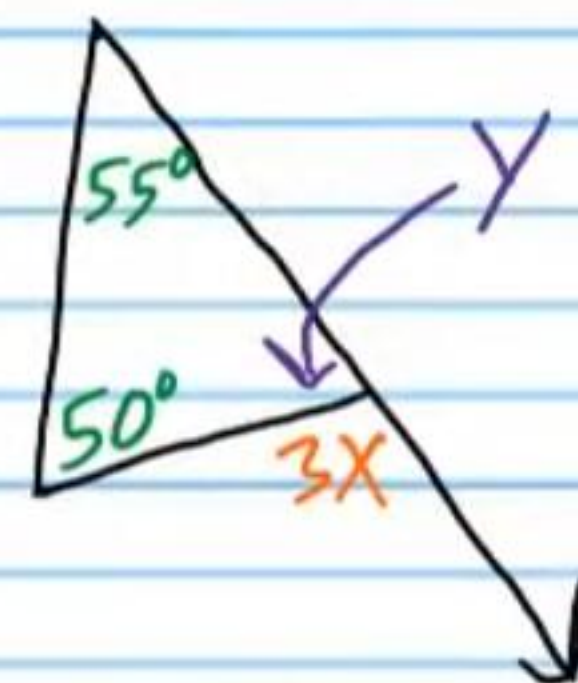
(11) i) If the following triangle were classified by its sides, what word would be used to describe it?



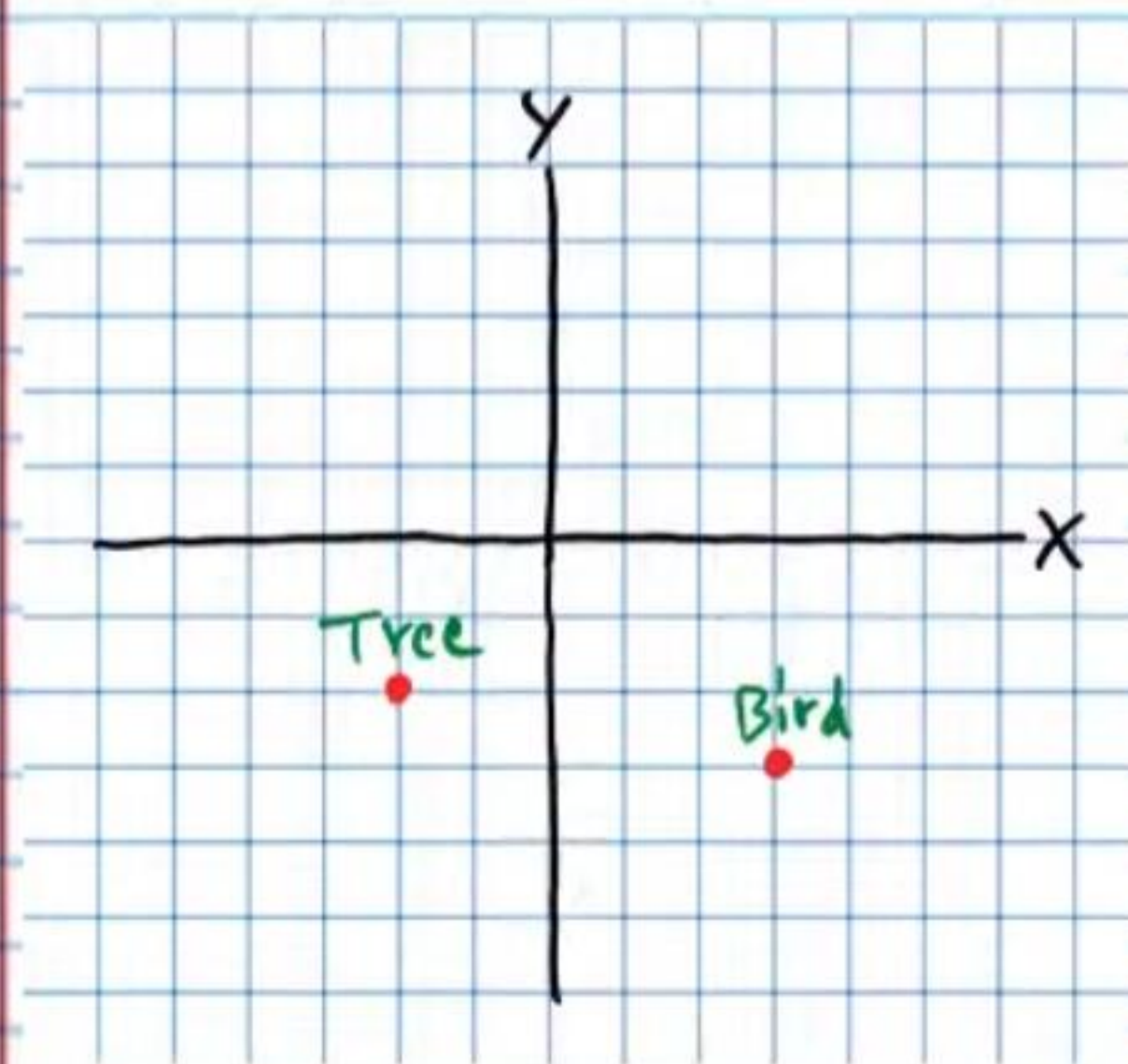
ii) If the triangle below were classified by its angles, what word would be used to describe it?



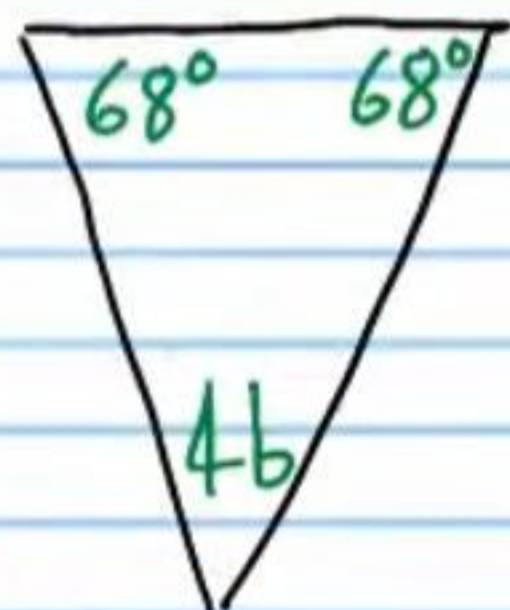
(12) Find x.



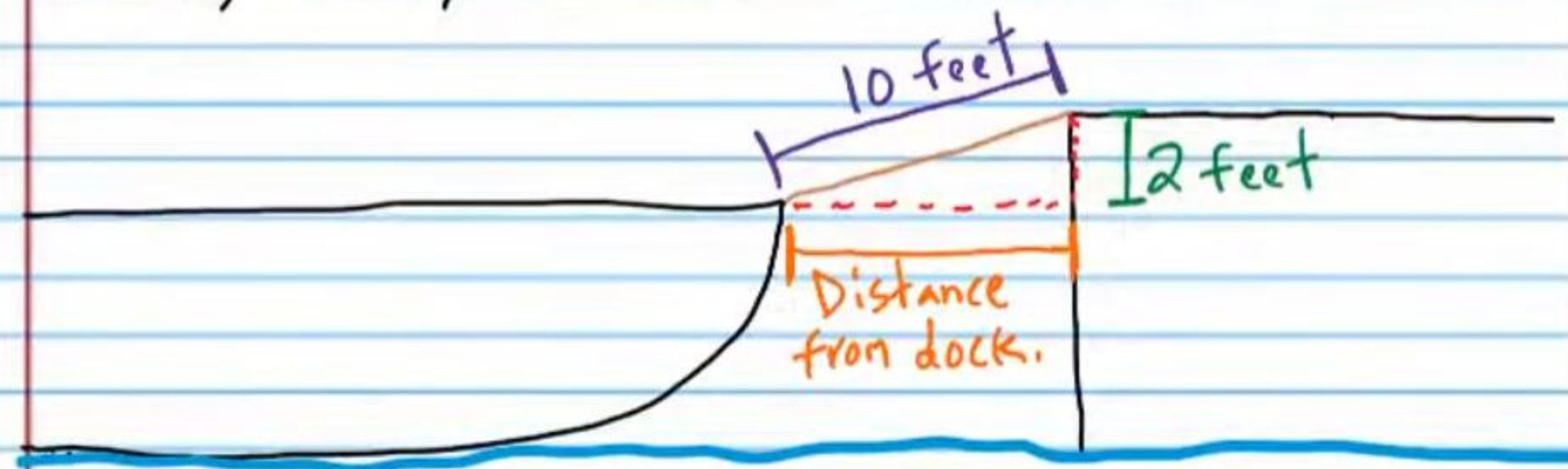
- ⑬ Find the distance between the bird and the tree. Each unit on the grid represents three meters.



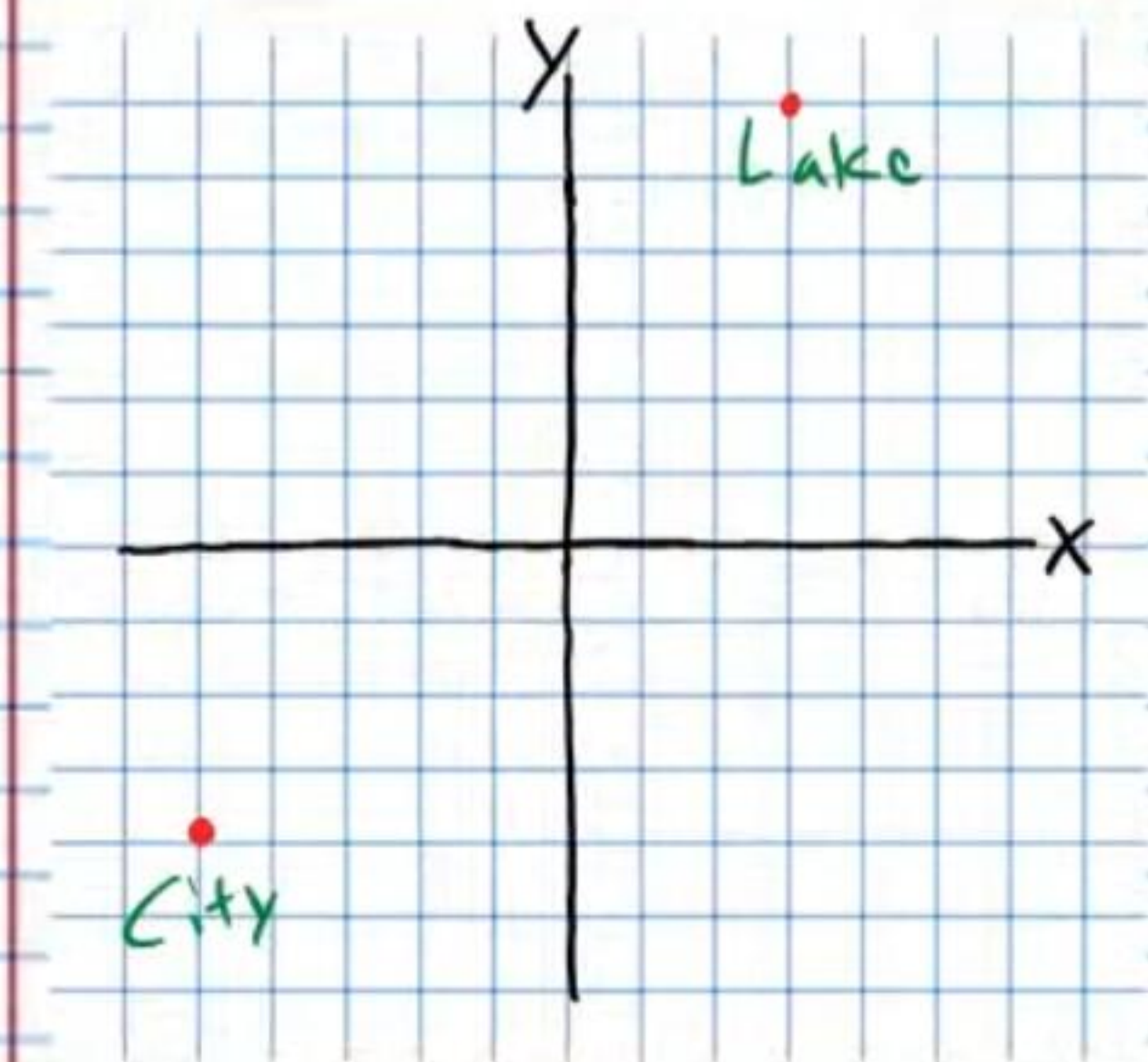
- ⑭ Find b .



- ⑮ A boat is tied to a dock. The dock is two feet above the boat and the rope is ten feet long. How far is the boat from the dock? You may leave your answer in root form.

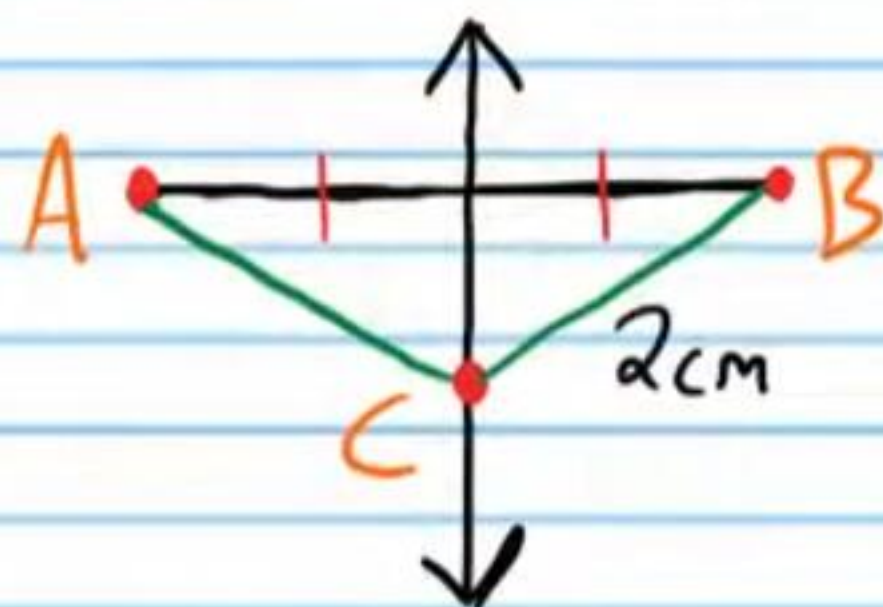


⑩ The hotel is halfway between the city and the lake. Find the coordinates of the hotel.

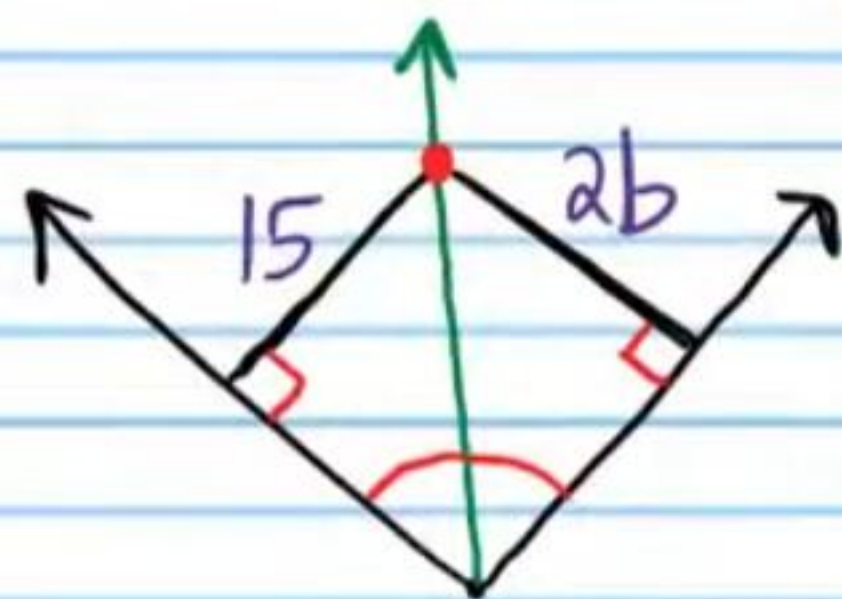


⑪ If the distance between $(X, 8)$ and $(2, -3)$ is 11, find the possible values of X .

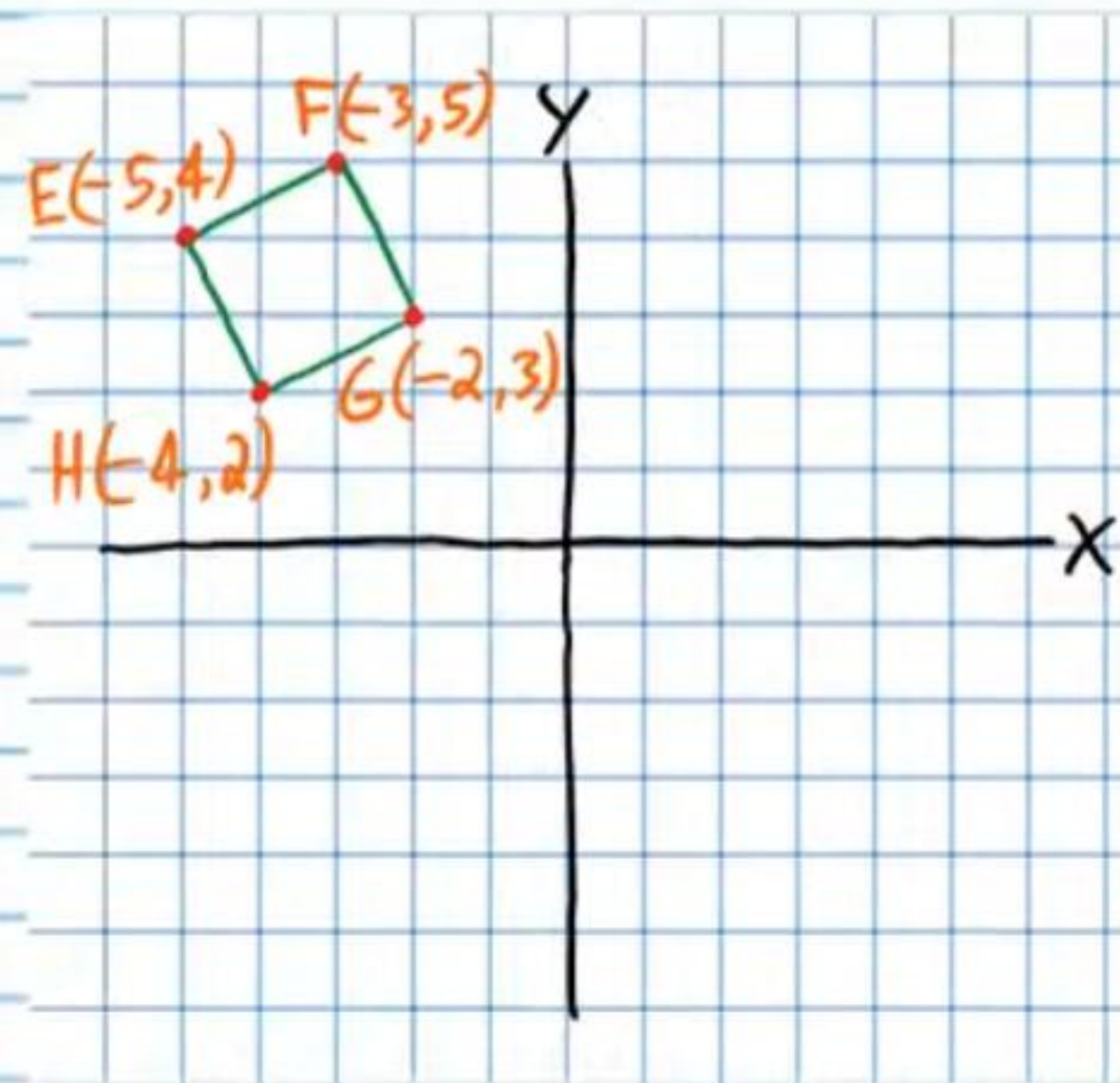
⑫ Find AC .



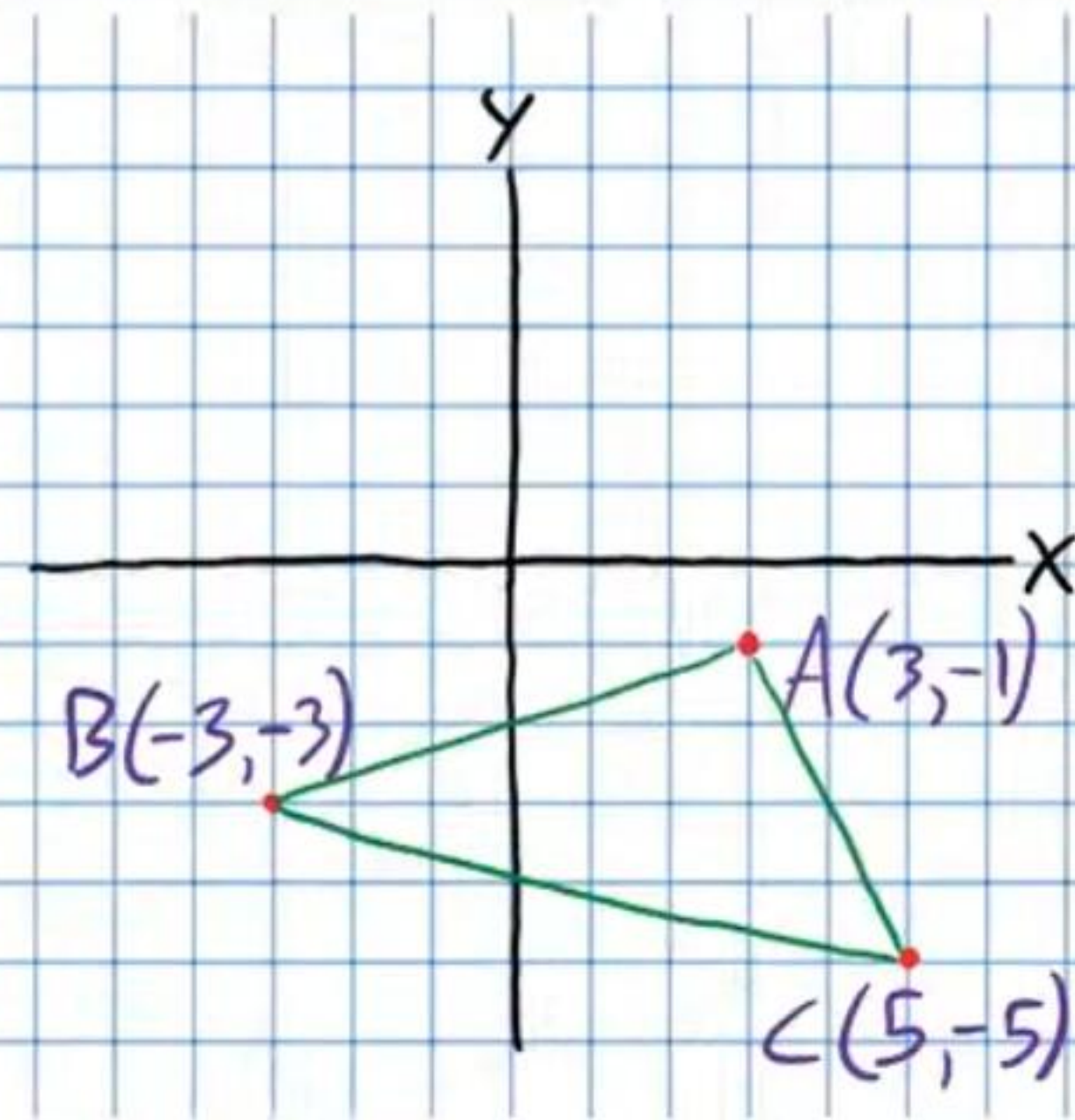
⑬ Find b .



Q20 Show that the quadrilateral below is a square.

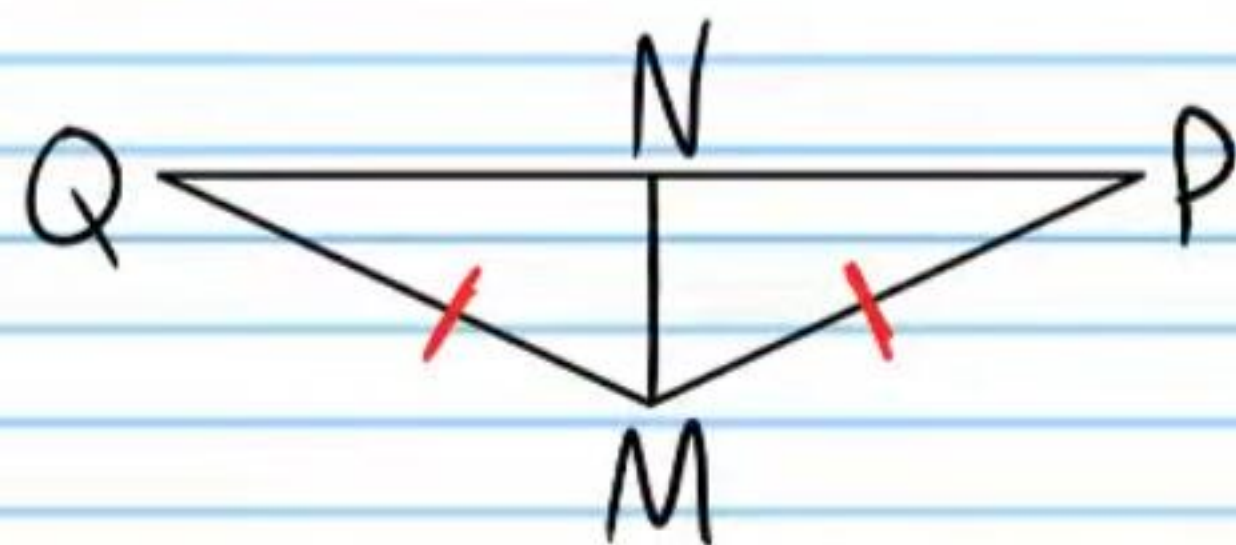


Q21 Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.



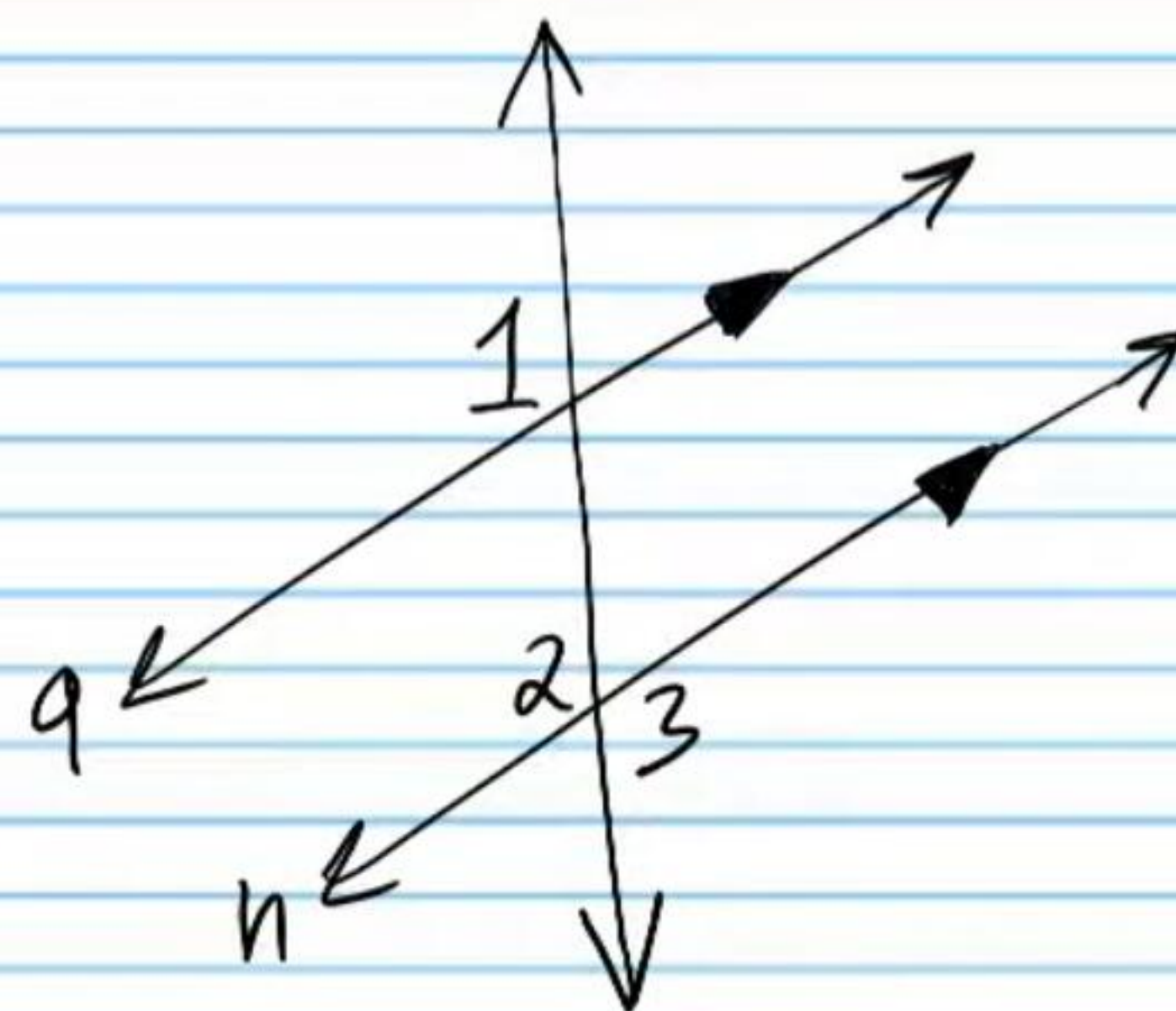
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

22) If $\overline{NM} \perp \overline{QP}$ and $\overline{QM} \cong \overline{MP}$, then $\triangle NQM \cong \triangle NPM$.



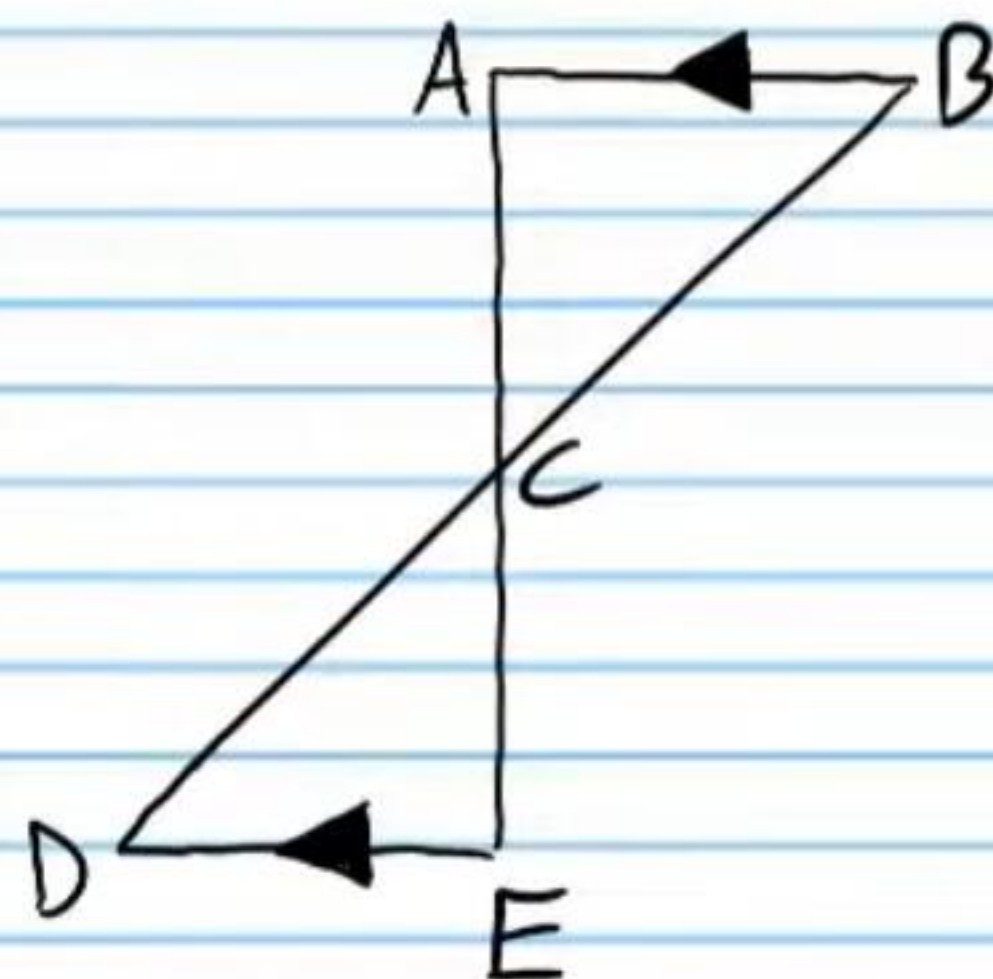
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

23) If $q \parallel n$ below, then $\angle 1 \cong \angle 3$. Note: This is the proof of the Alternate Exterior Angles Theorem. ★ You must use the Corresponding Angles Postulate and the Vertical Angles Theorem.



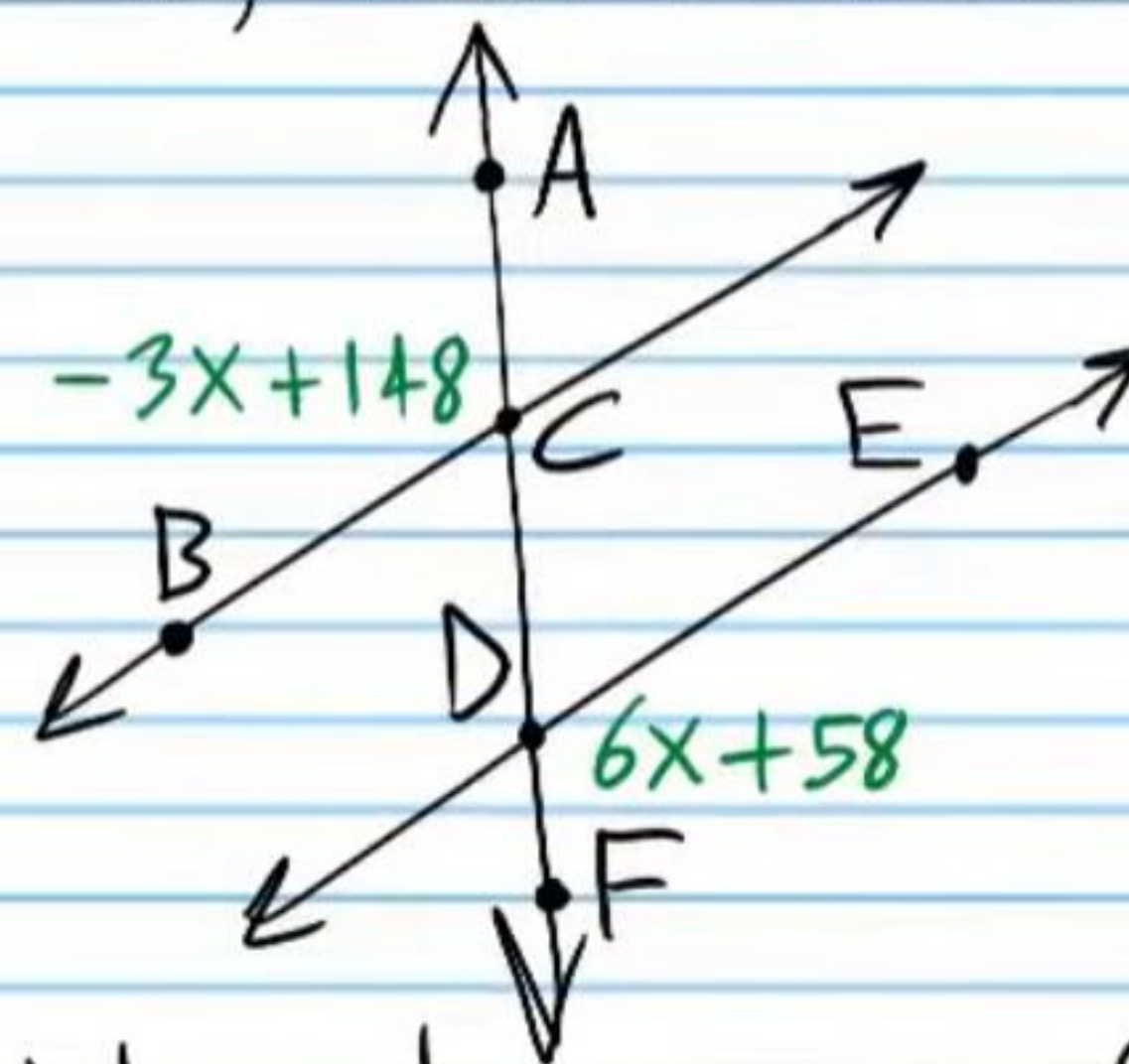
Statement	Reason
I)	
II)	
III)	
IV)	

24) If $\overline{AB} \parallel \overline{DE}$ and $\overline{AB} \cong \overline{DE}$, then C is the midpoint of $\overline{DB}$.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

25) If $x = 10$, then $\overline{BC} \parallel \overline{DE}$.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	
X)	

Practice Geometry Test II (Version 2) Answers

① Not enough Information

② $x = 74^\circ$, $\angle IACT$

③ $x = 7$

④ $b = 13$

⑤ $y = 5$

⑥ $(-980, -3)$

⑦ $m = \sqrt{11}$

⑧ i) $\triangle ABC \cong \triangle DEF$, SSS

ii) $\triangle ABC \cong \triangle ADC$, SAS

⑨ i) $\triangle ABC \cong \triangle DEF$, SSS

ii) $\triangle ABC \cong \triangle ADC$, ASA

⑩ i) Not enough information

ii) Not enough information

⑪ i) Isosceles

ii) Obtuse

⑫ $x = 35^\circ$

⑬ $3\sqrt{26} \text{ m}$

⑭ $b = 11$

⑮ $4\sqrt{6} \text{ ft}$

⑯ $(-1, 1)$

⑰ Not enough information

⑱ $b = 7.5$

⑲ $x = 2$

⑳ $M_{EF} = \frac{1}{2}$, $M_{FG} = -2$

$M_{EH} = -2$, $M_{HG} = \frac{1}{2}$

$D_{EF} = \sqrt{5}$, $D_{FG} = \sqrt{5}$, $D_{EH} = \sqrt{5}$, and $D_{HG} = \sqrt{5}$.

Therefore, the quadrilateral is a Square.

㉑ $M_{AB}(\frac{3-3}{2}, \frac{-3-1}{2}) = E(0, -2)$

$M_{CA}(\frac{3+5}{2}, \frac{-1-5}{2}) = F(4, -3)$

$M_{EF} = -\frac{1}{4}$, $M_{CB} = -\frac{1}{4}$

Therefore, $\overline{EF} \parallel \overline{CB}$.

$D_{EF} = \sqrt{(0-4)^2 + (-2-(-3))^2} = \sqrt{17}$

$D_{CB} = \sqrt{(-3-5)^2 + (-3-(-5))^2} = 2\sqrt{17}$

Therefore, $D_{EF} = \frac{1}{2} D_{CB}$

(22)	Statement	Reason
I)	$\overline{NM} \perp \overline{QP}$	Given
II)	$\angle QNM$ & $\angle PNM$ are rt. $\angle$'s.	Def. of $\perp$.
III)	$\overline{NM} \cong \overline{NM}$	Ref. Prop of $\cong$.
IV)	$\overline{QM} \cong \overline{MP}$	Given.
V)	$\triangle NQM$ and $\triangle NPM$ are right triangles.	Def. of a rt. $\triangle$.
VI)	$\triangle NQM \cong \triangle NPM$	HL

(23)	Statement	Reason
I)	$q \parallel n$	Given.
II)	$\angle 1 \cong \angle 2$	CAP
III)	$\angle 3 \cong \angle 2$	Vert. $\angle$'s Thm.
IV)	$\angle 1 \cong \angle 3$	Trans. Prop of $\cong$

(24)	Statement	Reason
I)	$\overline{AB} \parallel \overline{DE}$	Given.
II)	$\angle A \cong \angle E$ & $\angle D \cong \angle B$	AIAT

III)	$\overline{AB} \cong \overline{DE}$	Given.
IV)	$\triangle ABC \cong \triangle EDC$	ASA
V)	$\overline{BC} \cong \overline{DC}$	CPCTC
VI)	C is the midpoint of $\overline{DB}$.	Def. of midpoint.

You may choose to use the Vertical Angles Theorem and AAS instead.

(25)	Statement	Reason
I)	$x = 10$	Given.
II)	$m\angle ACB = -3x + 148$	Given (diagram)
III)	$m\angle ACB = -3(10) + 148$	Subs. Prop of $=$.
IV)	$m\angle ACB = 118^\circ$	Simplification
V)	$m\angle EDF = 6x + 58$	Given (diagram)
VI)	$m\angle EDF = 6(10) + 58$	Subs. Prop of $=$.
VII)	$m\angle EDF = 118^\circ$	Simplification.
VIII)	$m\angle ACB = m\angle EDF$	Trans. Prop of $=$.
IX)	$\angle ACB \cong \angle EDF$	Def. of $\cong \angle$'s.
X)	$\overline{BC} \parallel \overline{DE}$	AEACT